La Scuola d'Italia / Huson / Physics Nickname:
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2.1 Homework: Reference Frames and Displacement

Reference (provided). displacement = final position − starting position
distance = total length of the path traveled

1. On the grid, each square is 1 m, $+x$ is east, and $+y$ is north. Starting from the origin, draw and label each displacement vector.

   $\vec{A}$: 4 m east

   $\vec{B}$: 3 m north

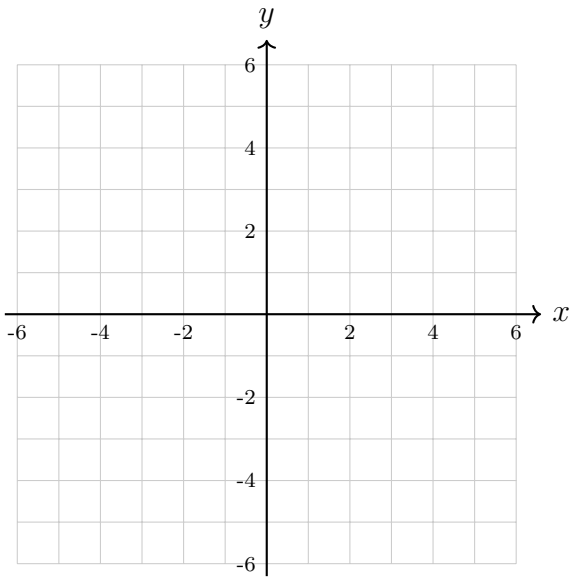
   $\vec{C}$: 5 m west

   $\vec{D}$: 2 m south

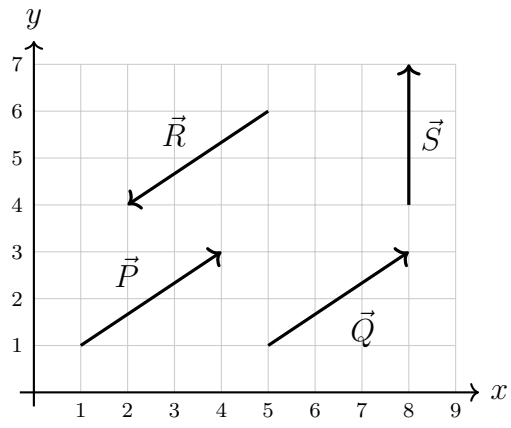
   $\vec{E}$: from the origin to $(-3, 4)$

   (a) $\vec{E}$ is _____ m west,
   and _____ m north.

   (b) How long is $\vec{E}$?



2. Four vectors are drawn on the grid below. A vector is set by its length and its direction only — it can be drawn anywhere.



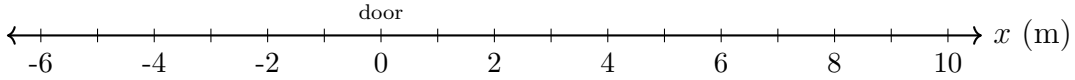
(a) Describe each vector.

	left/right	up/down
$\vec{P}$		
$\vec{Q}$		
$\vec{R}$		
$\vec{S}$		

(b) Which two vectors are equal?

(c) Which vector is the opposite of $\vec{P}$?

3. Take the classroom door as the origin, with east positive. A student walks 8 m east, then 5 m west, then 7 m west. Draw each displacement as an arrow above the number line, each one starting where the last one ended.



(a) Where does the student end up? $x =$ _____

(b) What is the student's displacement? _____

(c) What distance did the student walk? _____

4. Find the distance traveled and the displacement for each trip. Give a direction with each displacement that is not zero. A Manhattan block is about 80 m.

Trip	**Distance**	**Displacement**
(a) One lap of a 400 m running track		
(b) Empire State Building: 320 m up to floor 86		
(c) Up to floor 86, then back down to the lobby		
(d) Manhattan: 10 blocks north, then 4 blocks south		

5. **Reference frames.** A Frecciarossa train travels north at 300 km/h relative to the ground.

   (a) A passenger sits in their seat. What is their speed relative to the train? Relative to the ground?

   (b) The passenger walks at 5 km/h relative to the train. What is their velocity relative to the ground when they walk toward the front? Toward the back?

   (c) A train going south at 300 km/h passes on the other track. How fast does it rush past, as seen by a seated passenger?